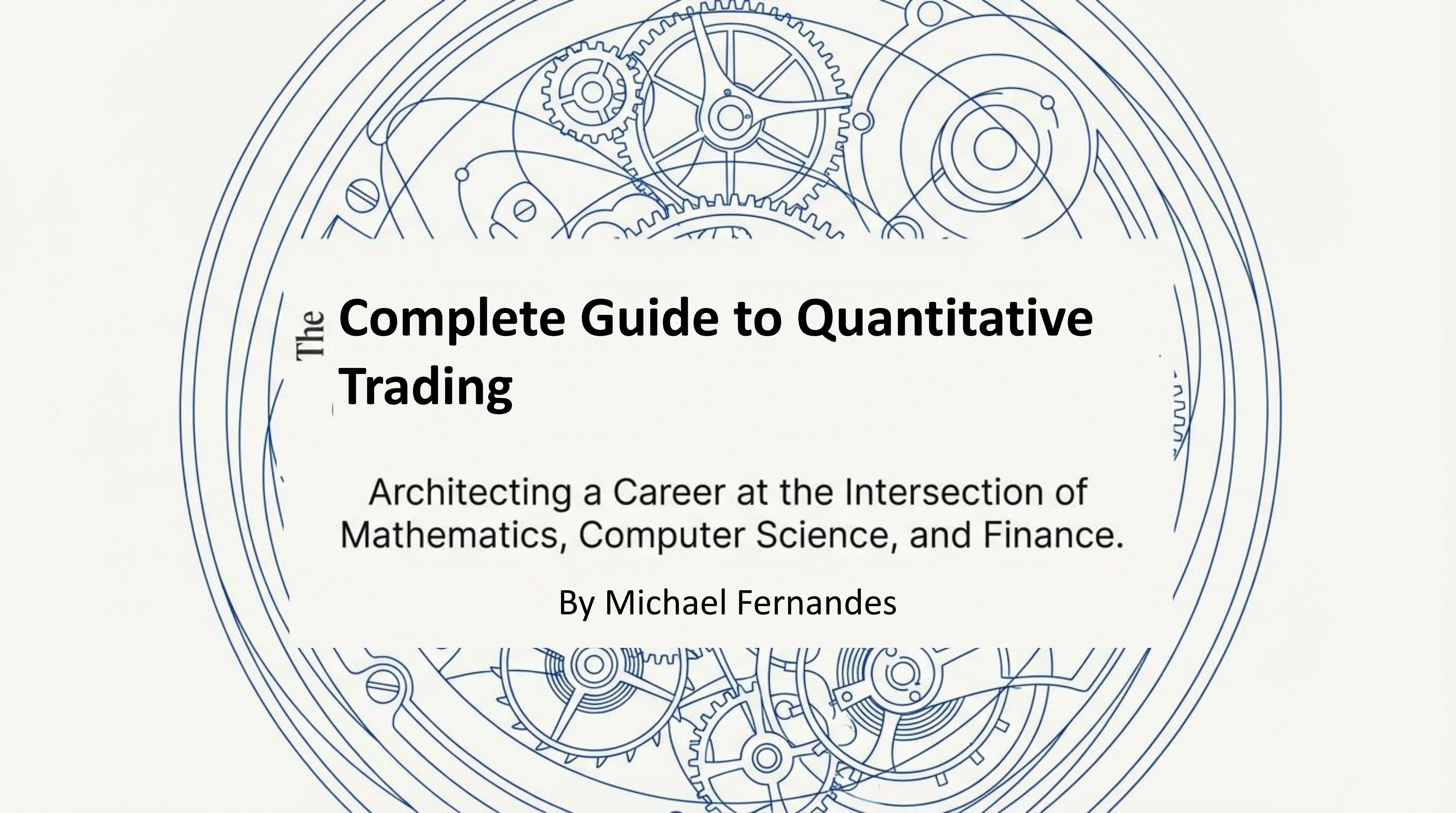




The Complete Guide to Quantitative Trading

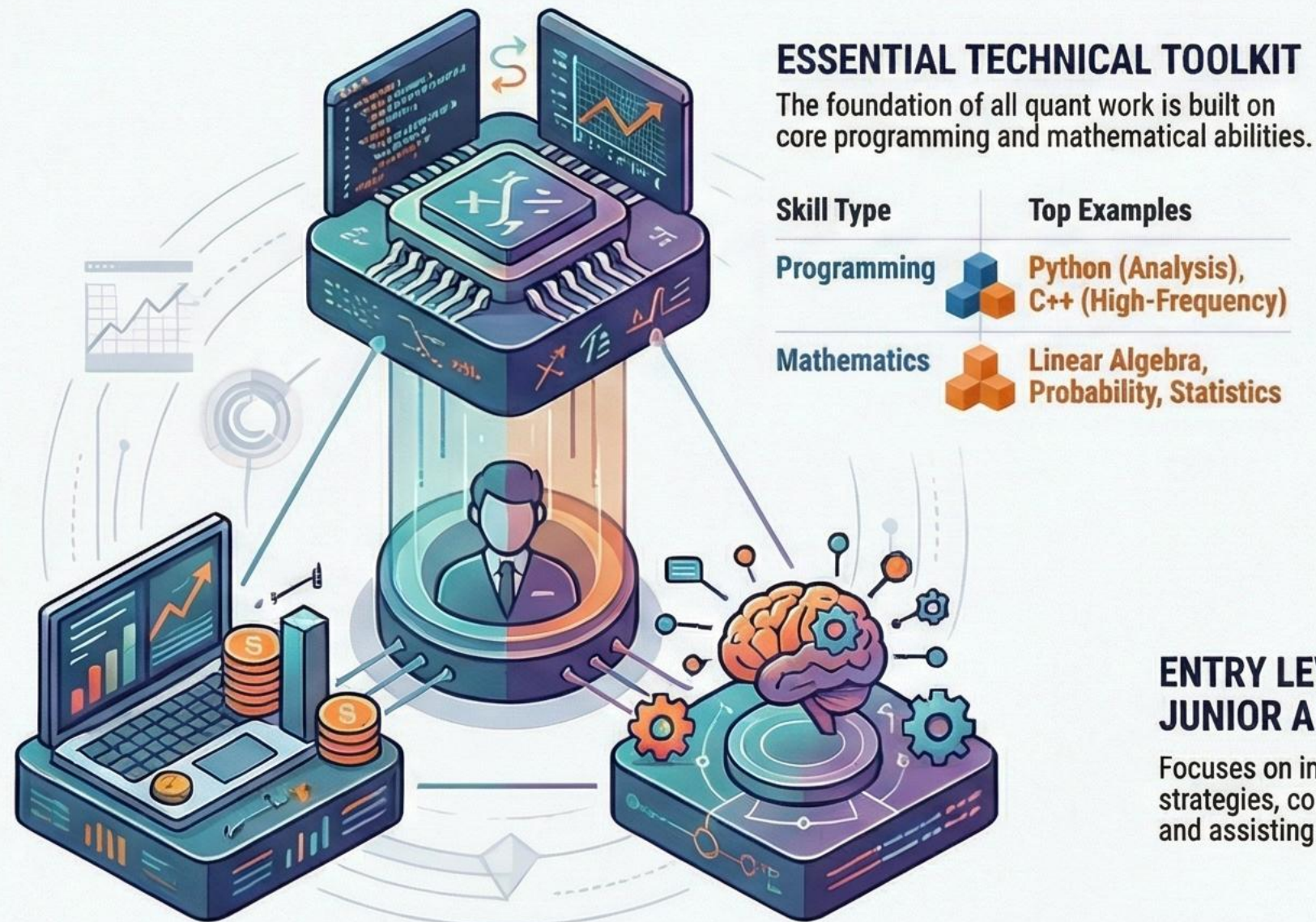
Architecting a Career at the Intersection of
Mathematics, Computer Science, and Finance.

By Michael Fernandes

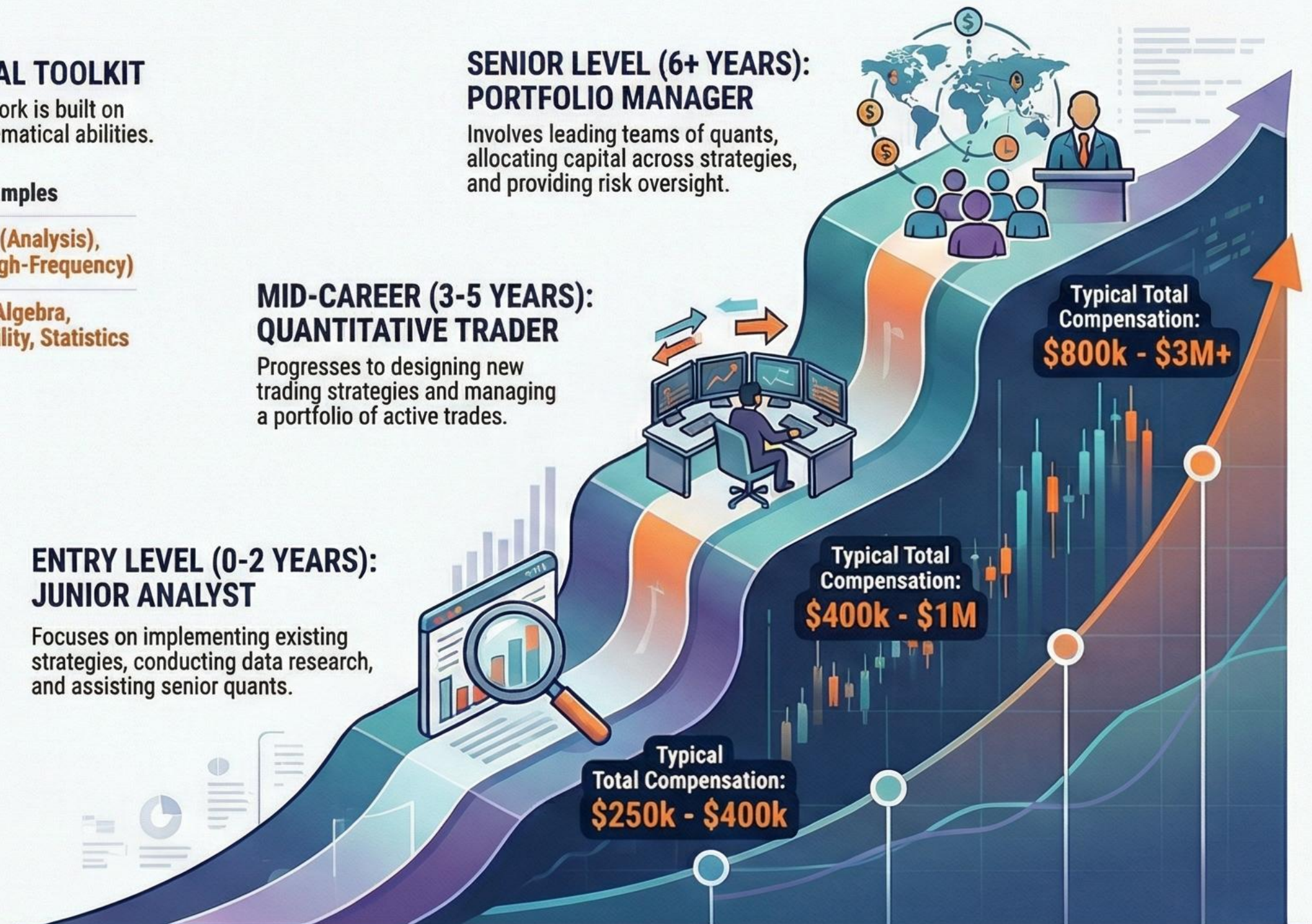
THE QUANTITATIVE TRADER'S BLUEPRINT

This infographic outlines the core competencies required to succeed and the typical career trajectory one can expect in this demanding but rewarding field.

THE ANATOMY OF A QUANT TRADER

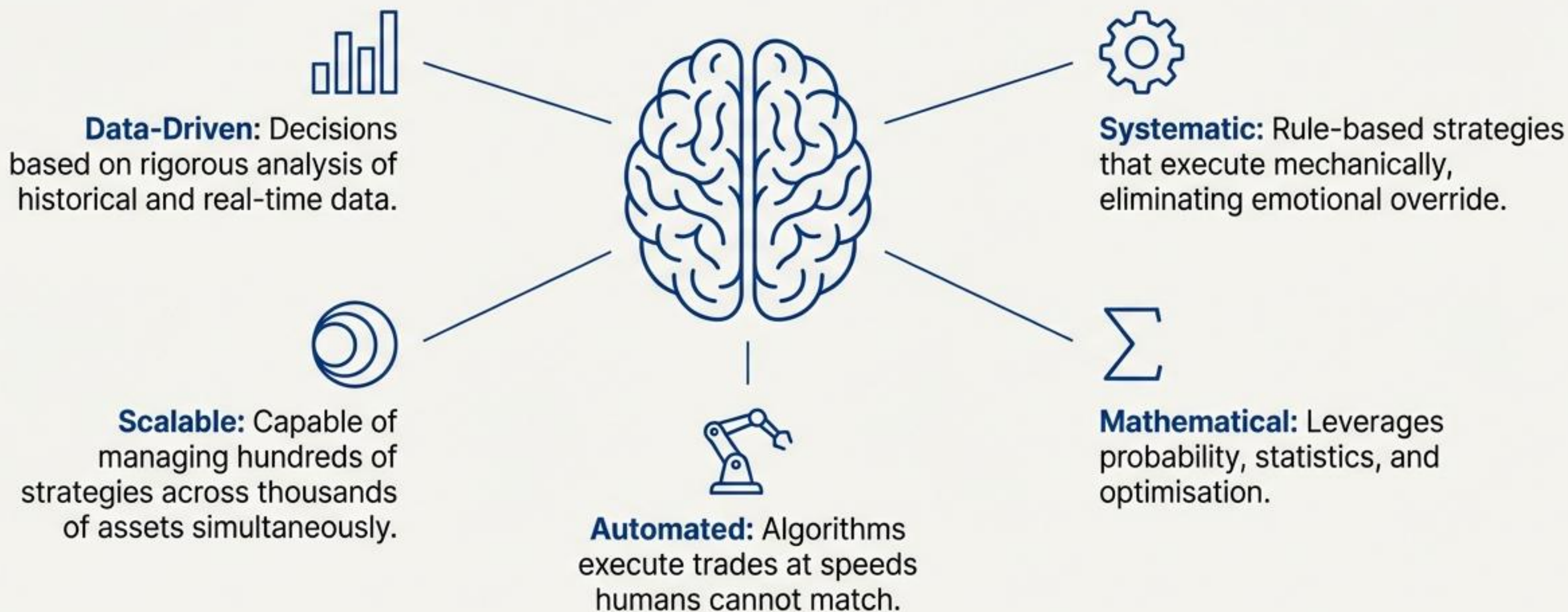


THE QUANT CAREER TRAJECTORY



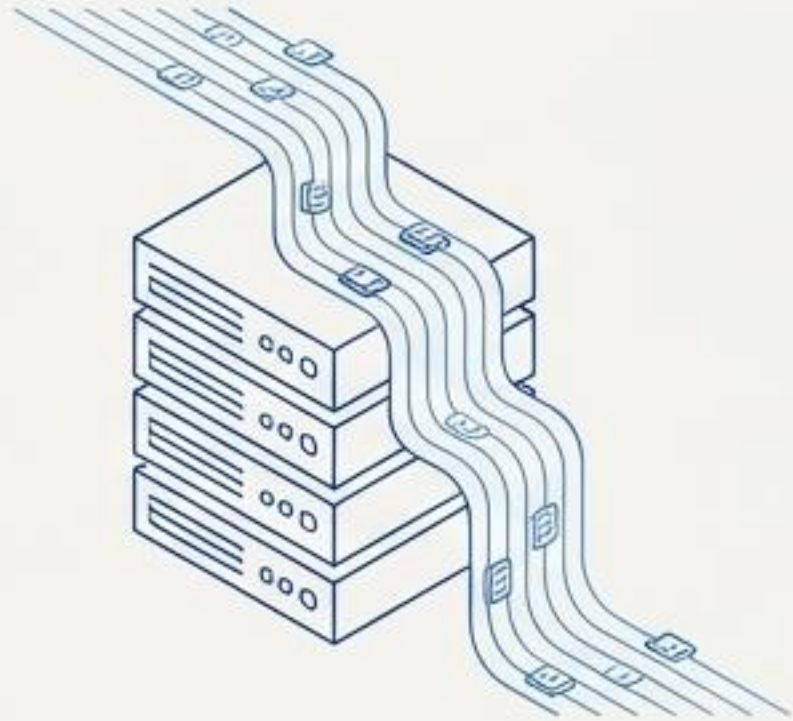
The Arena: Where Algorithms Replace Intuition

Quantitative trading uses mathematical models, statistical analysis, and computational algorithms to identify and execute trading strategies across financial markets.



The Two Philosophies: A Tale of Two Trading Desks

Quantitative Trading



Decision Making: Data-driven models
Speed: Milliseconds (HFT) to hours
Scale: Thousands of simultaneous trades
Emotion: Eliminated by design
Backtesting: Rigorous validation required
Scalability: Infinite (with capital)

Discretionary Trading



Decision Making: Trader judgment/experience
Speed: Minutes to days
Scale: Dozens of active positions
Emotion: Primary risk factor
Backtesting: Anecdotal historical review
Scalability: Limited by human attention

Forging the Quant: The Three Pillars of Essential Skill



Technical Skills
(The Foundation)

****Programming****: Python (for analysis), C++ (for HFT), SQL (for data).

Note on competency levels from Beginner to Advanced.



Mathematics & Statistics
(The Logic)

Core Maths: Linear Algebra, Probability Theory, Statistics, Calculus.

Advanced*: Stochastic Calculus (mentioning Itô's Lemma).



Financial Knowledge
(The Domain)

Core Finance: Options Pricing (Black-Scholes), Fixed Income, Market Microstructure.

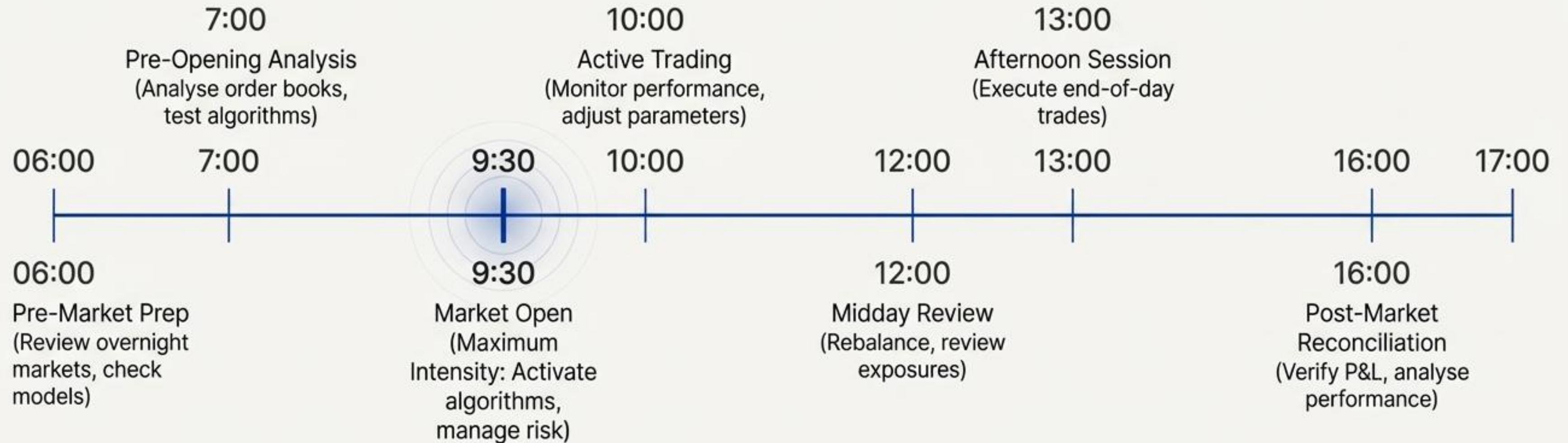
Soft skills are the glue: Problem-Solving, Attention to Detail, and Communication are critical for success.

The Ascent: From Junior Analyst to Portfolio Manager



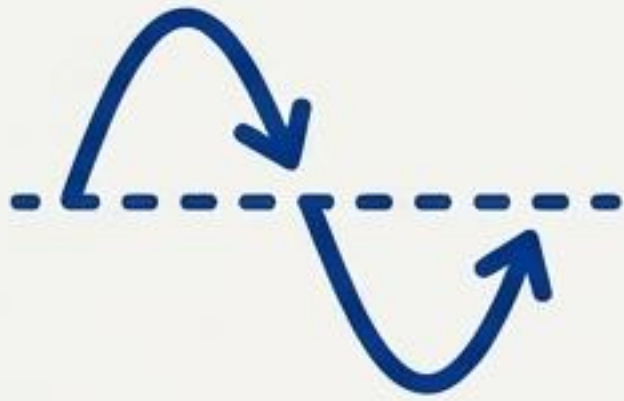
India Compensation: Path ranges from ₹12-20 LPA at entry to ₹1Cr+ at senior levels.

06:00 to 17:00: A Day on a Live Trading Desk



The day is a constant cycle of analysis, execution, and risk management, demanding intense focus and adaptability.

The Quant's Craft: Four Foundational Trading Strategies



Mean Reversion



Momentum



Statistical Arbitrage

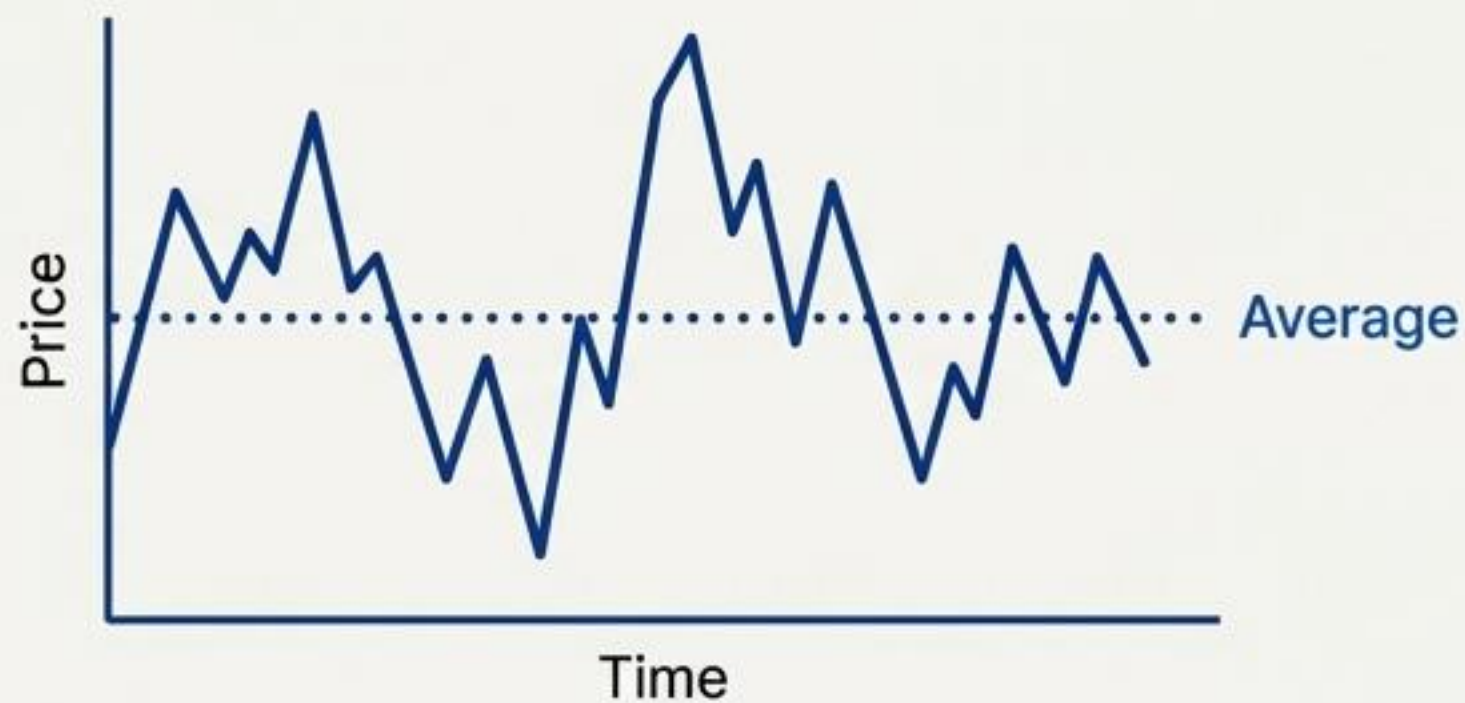


High-Frequency Trading

Each strategy exploits a different market inefficiency, from predictable human behaviour to microscopic latency advantages.

Betting on Extremes: Reverting to the Mean vs. Riding the Trend

Mean Reversion

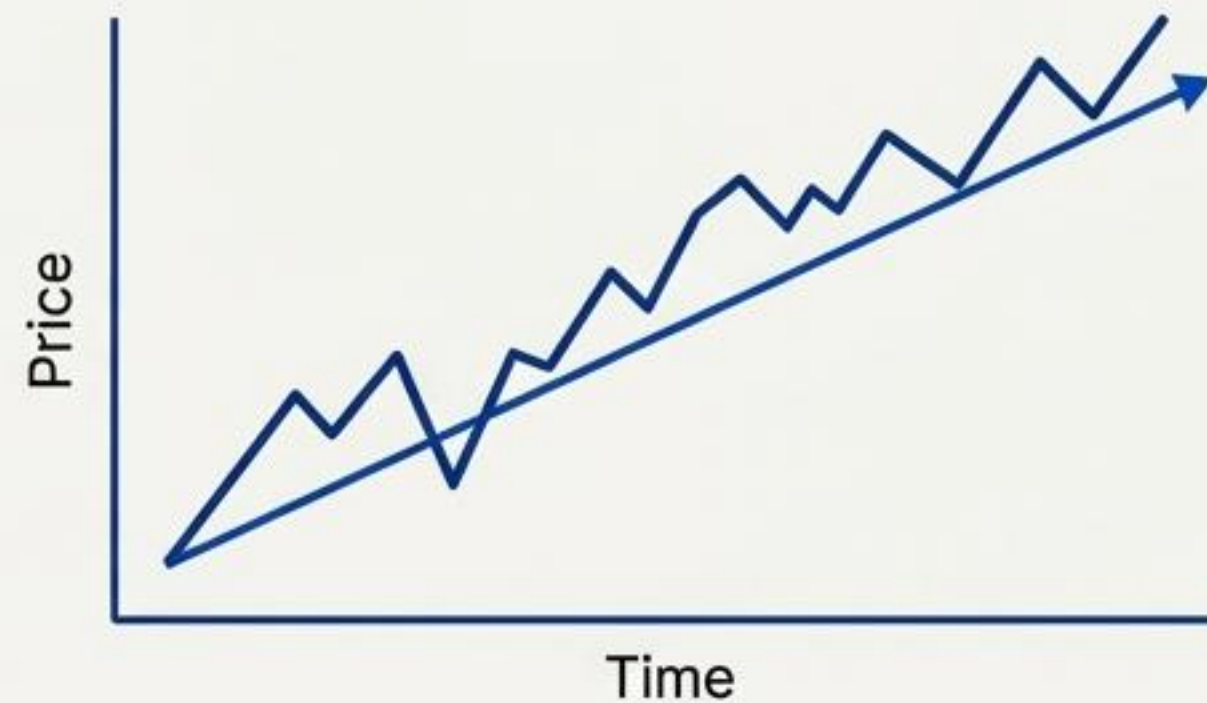


Concept: Assets that deviate from their average tend to revert back.

Example: "SPY trading at 432, well above its 200-day average of 428. A model would short SPY, expecting a 4-point drop."

Key Metrics: Win rate: ~62% (#006A4E green callout); Sharpe ratio improves 46% with VIX-based regime filtering.

Momentum



Concept: Assets with recent positive returns continue appreciating.

Example: "A stock with a +20% return over 3 months is likely to continue. A model would go long, riding the trend."

Key Advantage: Can capture large, extended market moves; higher Sharpe ratios in strong bull/bear regimes.

The Technological Edge: Arbitrage and High-Frequency Trading

Statistical Arbitrage



Concept: Exploit temporary mispricings between historically correlated securities.

Mechanics: Identify a correlated pair (e.g., Stock A & B). When their price spread widens abnormally, long the cheaper asset and short the more expensive one, betting the spread will normalise.

Risk: Correlation breaks, and the spread continues to widen.

High-Frequency Trading (HFT)

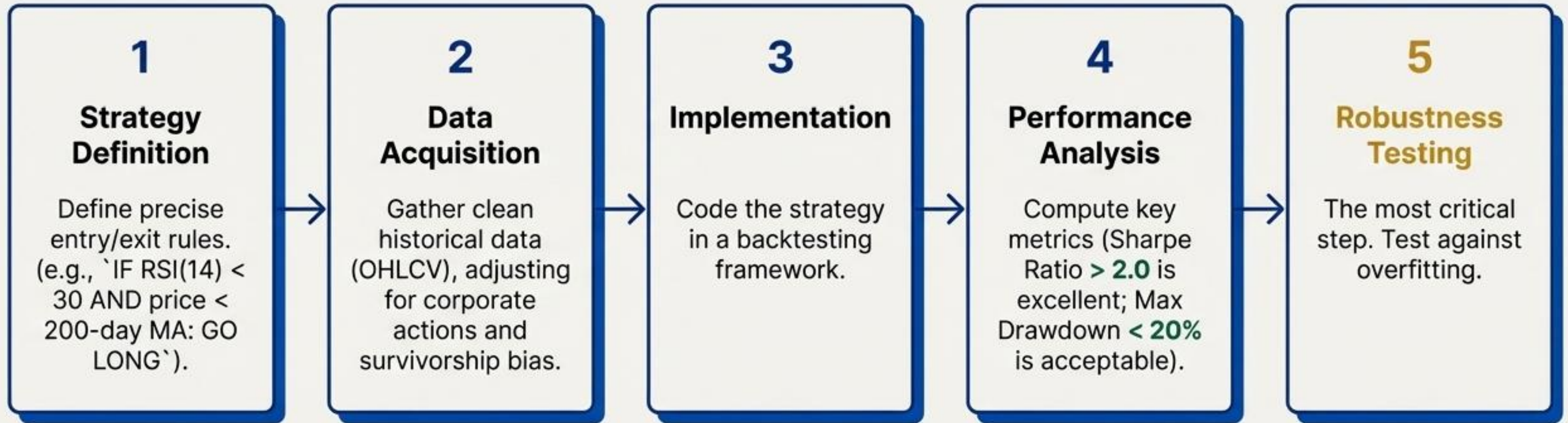


Economics: Profit = Tiny Edge (~0.1-0.5 basis points) × Enormous Volume (millions of trades daily).

Technology is Everything: Co-location, Direct Market Access (DMA), Custom FPGA/GPU chips.

Latency is King: Market-leading HFT firms target <100 microseconds latency.

The Workshop: A Five-Step Workflow for Strategy Development



A good backtest is hard to build. A great backtest is one that tells you not to trade.

The Code Behind the Craft: Backtrader vs. VectorBT

Backtrader (Beginner-Friendly)

```
import backtrader as bt

class RSIStrategy(bt.Strategy):
    def __init__(self):
        self.rsi = bt.indicators.RSI(self.data.close)

    def next(self):
        if self.rsi[0] < 30:
            self.buy()
        elif self.rsi[0] > 70:
            self.close()
```

VectorBT (Performance-Focused)

```
import vectorbt as vbt
import pandas as pd

# Load data
close = data['Close']

# Define strategy (vectorized)
entries = rsi < 30
exits = rsi > 70

# Backtest (extremely fast)
portfolio = vbt.Portfolio.from_signals(close, entries, exits)
```

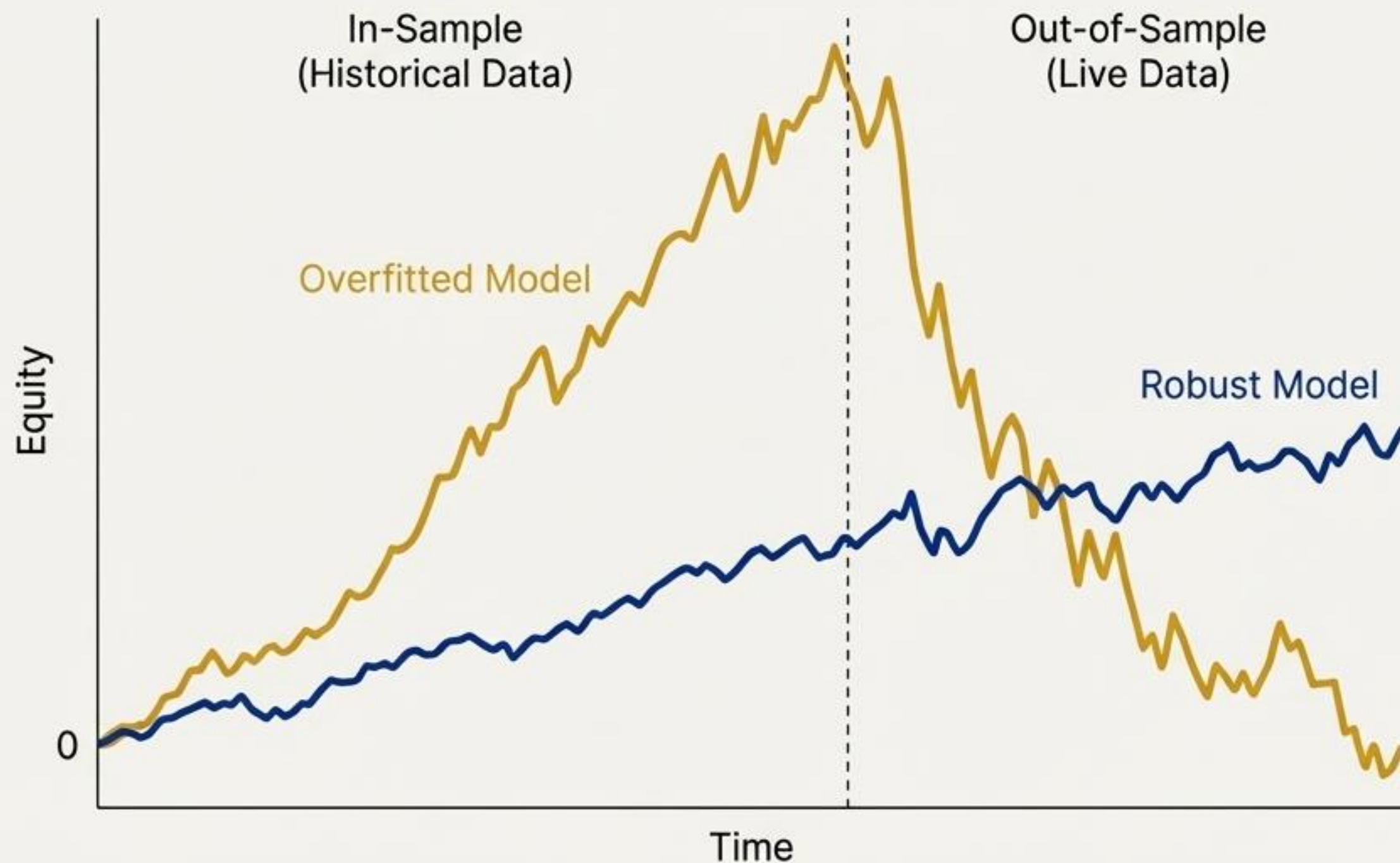
Performance Showdown

VectorBT: 10,000 iterations in **5 seconds**
Backtrader: 10,000 iterations in **2+ minutes**

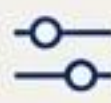
Conclusion: VectorBT is **20-40x** faster due to vectorization, enabling far more extensive parameter testing.

The Siren Song of the Backtest: Avoiding the Overfitting Trap

Overfitting occurs when a model is optimised so perfectly to historical data that it fails catastrophically on new, live data.



Checklist: Key Prevention Techniques

-  Walk-Forward Testing (rolling training/testing windows)
-  Out-of-Sample Testing (hold out 20% of data for validation)
-  Parameter Sensitivity Analysis (test $\pm 10\%$ variations)
-  Monte Carlo Simulations (randomise trade sequences)

Red Flags

- Backtest Sharpe of 2.5 vs. Out-of-Sample Sharpe of 0.8
- Parameters that must be exact (e.g., 73.2 days, not 73 or 74)

The Gauntlet: Surviving the Four Stages of the Quant Interview

CV Screening

Filter: GPA > 3.5, relevant projects, programming skill.

Acceptance Rate: 5-10% of applicants advance



Online Assessment

Test: Timed mental math (e.g., 80 questions in 8 mins), logic puzzles.

Cutoff: Top 20% on speed and accuracy advance



HR / Motivational Interview

Focus: 'Why quant?', 'Why our firm?', problem-solving (STAR method).

Advancement Rate: ~60% advance



The Technical Interviews (The Main Event)

Format: 1-2 hours, back-to-back with traders, high pressure.

Content: Probability, brainteasers, trading scenarios.



In the Hot Seat: Classic Quant Interview Questions

Probability & Expected Value



Q: "You flip a coin until you get heads. What's the expected number of flips?"

A: " $E = 1 \times (1/2) + 2 \times (1/4) + 3 \times (1/8) + \dots = 2$ flips"

Trading Scenarios



Q: "The Fed just announced a rate hike. What happens to the stock market?"

A: "Nuanced answer required: short-term negative (discount rates), but potentially positive if it confirms a strong economy. Depends on what was priced in."

Brainteasers / Logic



Q: "How many piano tuners are in New York?"

A: "Fermi estimation: ~8M people -> ~8,000 pianos -> ~16,000 tunings/year. Tuner does ~1,000/year -> ~16 tuners."

Mental Math



Q: " $47 \times 23 = ?$ "

A: " $(40 \times 20) + (40 \times 3) + (7 \times 20) + (7 \times 3) = 800 + 120 + 140 + 21 = 1,081$. Target: <5 seconds."

Edge is Everything: The Reality of the Arena

The Hard Truths

- Most strategies (90%+) fail in live trading.
 - Competition is intense, featuring the best from MIT, Stanford, etc.
 - Psychological pressure is immense; losing \$100k in a day is part of the job.
 - Strategies decay rapidly; constant innovation is mandatory, not optional.
-

The Three Pillars of Sustainable Edge



1. Superior Data: Proprietary or alternative datasets competitors lack.



2. Superior Modelling: Advanced ML, regime detection, and ensemble methods.



3. Superior Execution: Lowest costs and fastest latency.

The market has zero sympathy for pretenders. Only those with a proven, defensible edge survive.